**THE PRIME MINISTER'S OFFICE**
**REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT**
**LONGIDO DISTRICT MID TERM JOINT EXAMINATION**

**FORM THREE**
**032 CHEMISTRY**

**Time: 3 Hours** **Thursday, 27th August 2026 a.m.**

## Instructions

1. This paper consists of Sections A, B and C with a total of eleven questions.
2. Answer **all** questions in Sections A and B and **two** questions from Section C.
3. Section A carries **16 marks**, Section B carries **54 marks**, and Section C carries **30 marks**.
4. All calculations must show clear working steps.
5. Use the following atomic masses where necessary: H = 1, C = 12, N = 14, O = 16, Na = 23, Mg = 24, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Cu = 64, Zn = 65.

## SECTION A (16 MARKS)

Answer **all** questions in this section.

1. For each item (i)–(x), choose the correct answer from among the given alternatives and write its letter in the answer booklet provided.

(i) A student observes that perfume sprayed at one end of a classroom can eventually be detected at the other end. This illustrates:
A. Freezing
B. Diffusion
C. Melting
D. Sedimentation
E. Osmosis

(ii) During the preparation of oxygen in the laboratory, the first portion of gas is discarded mainly because:
A. oxygen is poisonous
B. it contains air initially present in the apparatus
C. it contains only carbon dioxide
D. oxygen cannot support combustion initially
E. the reaction first produces hydrogen

(iii) Which observation provides the strongest evidence that a chemical change has occurred?
A. Ice changes into water
B. Salt dissolves in water
C. A gas is produced when two substances react
D. Water changes into steam
E. Paper is cut into pieces

(iv) Which of the following substances is an acid?

A. Sodium hydroxide
B. Calcium oxide
C. Hydrochloric acid
D. Sodium chloride
E. Magnesium oxide

(v) A solution has a pH of 2. Which statement correctly describes the solution?
A. It is strongly alkaline.
B. It is weakly alkaline.
C. It is neutral.
D. It is strongly acidic.
E. It contains no ions.

(vi) Which pair represents a temporary hard water treatment method and a permanent hard water treatment method respectively?
A. Filtration and decantation
B. Boiling and addition of washing soda
C. Distillation and filtration
D. Evaporation and boiling
E. Addition of soap and filtration

(vii) Which equation represents a neutralisation reaction?
A. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B. $CaCO_3 \rightarrow CaO + CO_2$
C. $HCl + NaOH \rightarrow NaCl + H_2O$
D. $C + O_2 \rightarrow CO_2$
E. $Mg + O_2 \rightarrow MgO$

(viii) How many moles are present in 5.85 g of sodium chloride, NaCl?
A. 0.01 mol
B. 0.05 mol
C. 0.10 mol
D. 0.50 mol
E. 1.00 mol

(ix) Which apparatus is most appropriate for transferring an accurately measured fixed volume of solution during titration?
A. Beaker
B. Measuring cylinder
C. Pipette
D. Test tube
E. Evaporating dish

(x) An indicator is used in an acid-base titration mainly to:
A. increase the concentration of the acid
B. provide heat for the reaction
C. show when the reaction has reached its end point
D. separate the acid from the base
E. increase the volume of the solution

2. Match each item in **List A** with the most appropriate response in **List B** by writing the letter of the correct response beside the item number.

| LIST A | LIST B |
|---|---|
| (i) A substance that changes colour according to the acidity or alkalinity of a solution.<br>(ii) A solution whose concentration is accurately known.<br>(iii) Water that does not readily form lather with soap because it contains dissolved calcium or magnesium ions.<br>(iv) The process of converting a soluble salt into a pure solid form by removing solvent.<br>(v) A chemical substance that reacts with an acid to form salt and water.<br>(vi) The amount of substance containing $6.02\times10^{23}$ particles. | A. Base<br>B. Indicator<br>C. Hard water<br>D. Evaporation<br>E. Standard solution<br>F. Mole<br>G. Filtration<br>H. Catalyst |

## SECTION B (54 MARKS)

Answer **all** questions in this section.

3. (a) An aqueous solution of calcium hydroxide is reacted with dilute hydrochloric acid.
   (i) Write the balanced chemical equation for the reaction.
   (ii) Identify the type of chemical reaction taking place.
   (iii) Name the salt formed.
   (b) A student adds dilute hydrochloric acid to sodium carbonate solution.
   (i) State two observations expected during the reaction.
   (ii) Write the balanced chemical equation for the reaction.
   (iii) Name the gas produced.

4. Elements X, Y and Z have the following electronic configurations:
   X = 2:8:1
   Y = 2:8:7
   Z = 2:8:8
   (i) State the group and period of elements X and Y .
   (ii) Which element is most likely to lose one electron during chemical combination?
   (iii) Which element is most likely to gain one electron?
   (iv) Explain why element Z is chemically unreactive.
   (v) State the type of bond formed between X and Y.
   (vi) Explain briefly how the bond is formed.

5. A student determines the concentration of sodium hydroxide solution by titrating it against 0.100 mol/dm³ hydrochloric acid. The student pipettes 25.0 cm³ of the sodium hydroxidesolution into a conical flask. The average titre of hydrochloric acid is 20.0 cm³.
   (i) Briefly give the meaning of a 'Standard solution' and an 'End point of a titration'.
   (ii) Calculate the number of moles of HCl used in the titration.
   (iii) Calculate the number of moles of NaOH that reacted.
   (iv) Calculate the concentration of the NaOH solution in mol/dm³.

6. Water obtained from two different sources is tested with soap. Sample **A** forms lather readily, whereas Sample **B** produces scum and requires more soap before lather forms.
   (i) Identify Sample A and Sample B as either soft water or hard water.
   (ii) Explain how temporary hardness can be removed by boiling, including a suitable chemical equation.

(iii) Explain how washing soda can be used to remove permanent hardness from water.
(iv) State two disadvantages of hard water in domestic or industrial activities.

7. A student reacts 4.8 g of magnesium completely with excess hydrochloric acid. Calculate:
(i) The number of moles of magnesium used.
(ii) The volume of hydrogen gas produced at STP.
(iii) State two observations that would be made during the reaction.
(iv) Identify the substance that is in excess in this experiment and explain what this means.

8. (a) A Form Three student wants to investigate whether increasing temperature affects the rate at which sugar dissolves in water. State the following:
(i) The aim of the investigation.
(ii) The independent variable.
(iii) The dependent variable.
(b) Outline at least six steps the student should follow when carrying out the investigation scientifically.

## SECTION C (30 MARKS)

Answer any **two** questions from this section.

9. Currently, Tanzania is suffering from climatic change disaster which is caused by over dependency on forests as a source of fuel and over production of waste material that pollute our environment. To overcome this problem, the use of renewable sources of energy that does not release harmful gases to the atmosphere is recommended. As a Chemistry student, prepare a report to convince the community to use the renewable sources of energy while they are conserving the environment.

10. Seypalm heated **10.0 g** sample of calcium carbonate until the reaction was complete, where a gas that turns lime water milky was released.
(a) Calculate the relative formula mass of calcium carbonate.
(b) Calculate the number of moles of calcium carbonate that was initially present. Determine the number of moles of the evolved gas.
(c) Calculate the volume of the evolved gas at room temperature and pressure.
(d) Calculate the mass of the evolved gas.
(e) State two observations that would show that the reaction is complete.

11. (a) A student is given a mixture containing iron filings, sand and sodium chloride.
(i) Classify the mixture as homogeneous or heterogeneous. Give a reason.
(ii) Describe a step-by-step procedure for separating the three substances and recovering each one in a dry form.
(b) A water sample contains dissolved calcium hydrogencarbonate.
(i) Name the type of hardness produced by this salt.
(ii) Explain how boiling removes this type of hardness, giving a balanced equation.
(c) State three differences between physical and chemical changes.